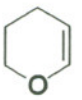
ORGANIC CUMULATIVE EXAM
DECEMBER 7, 2001

The reference for a reactivity study of norbornyne was posted last week: Laird, D. W.; Gilbert, J. C. *J. Am. Chem. Soc.* **2001**, *123*, 6704-6705. Answer the following questions.

1. What originally prompted this work, e.g. what results in earlier referenced work led the authors to pursue their study? In Woodward-Hoffman terms, was the reaction observed in this earlier work allowed or forbidden? Explain.

2. Describe two methods for generating norbornyne. Give the mechanisms of these reactions.

3. Give the four products that are formed when norbornyne is generated in the presence of 3,4-dihydro-2H-pyran (shown below).



4. Give the mechanisms for the formation of the four products given in question 3. Explain the relevant interaction between the molecular orbitals on norbornyne and the dihydropyran.

5. Give precedents that were outlined in this communication for any of the mechanistic steps.